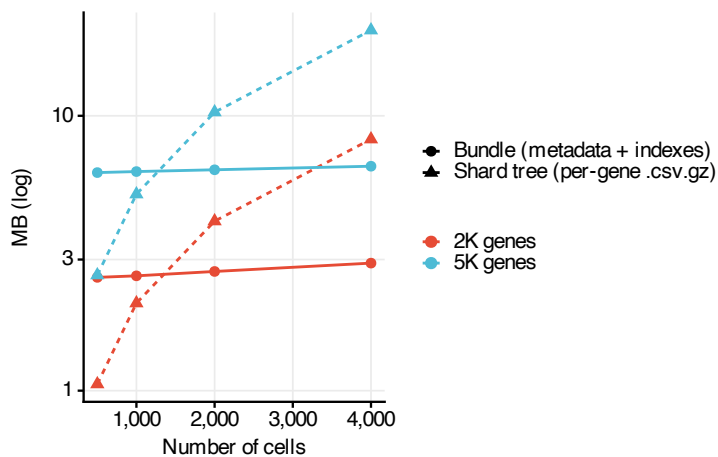
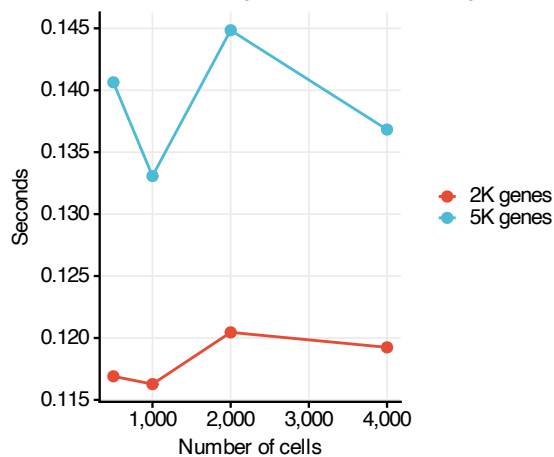
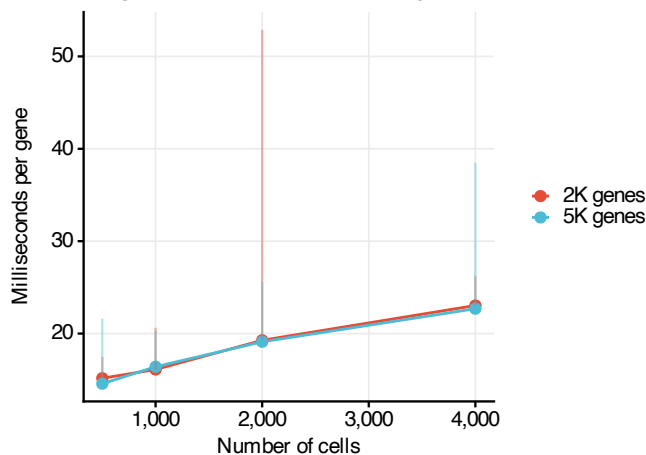
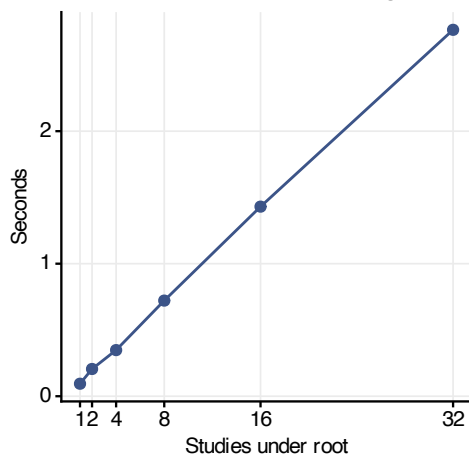
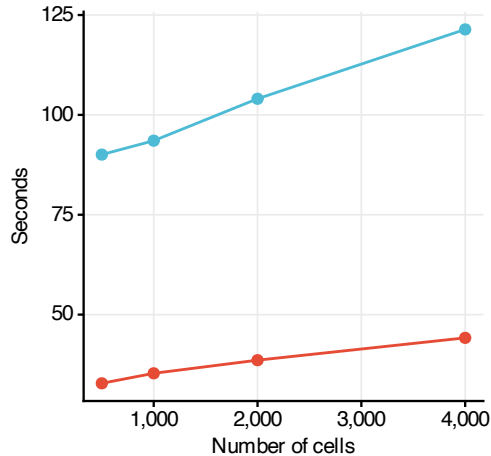


A. Bundle size vs shard-tree size**B. load_study() cold-start latency****C. gene_values() fetch latency (median, bar to max)****D. discover_studies() scaling****E. prepare_study_data() wall time****F. prepare_study_data() peak memory**